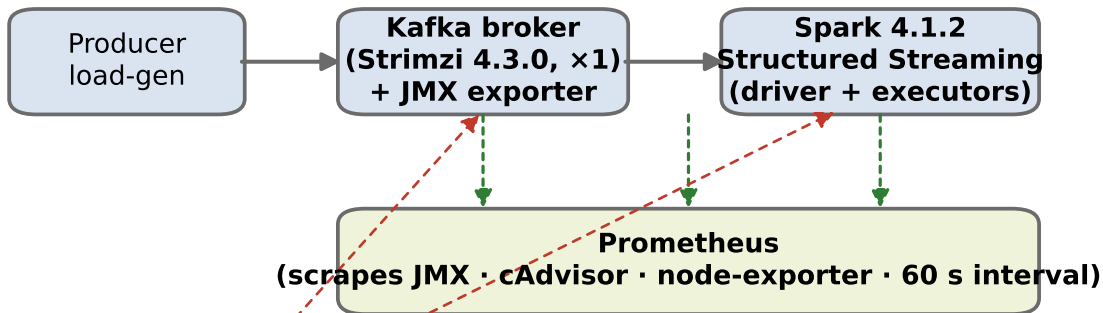


Kind single-node Kubernetes cluster



Fault injection
(`fault_injection/*.py`)
real, physical faults

targets:

- `broker_kill` → Kafka
- `executor_oom` → Spark exec
- `disk_pressure` → node fs
- `network_degradation` → broker net
- `backpressure_cascade` → Spark streaming (excluded: no independent metric)